

Large Language Models - Exam

Politecnico di Torino

27 Giugno 2025

Tempo impiegato: 1 ora 14 min. **Valutazione:** 11,80 / 18,00

Domanda 1 (2 points (no penalty for a wrong answer))

The attention for matrices Q, K, V (queries, keys, values) is computed as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Where d_k is the dimensionality of the keys.

The following are the two input tokens that reach an attention layer:

$$x = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix}$$

The following are the W_q, W_k, W_v matrices for the attention layer:

$$W_q = \begin{bmatrix} -2 & -1 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}, \quad W_k = \begin{bmatrix} 0 & -2 \\ -1 & 0 \\ -1 & 1 \end{bmatrix}, \quad W_v = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Compute the output of the attention operation for the **first token**. Write the answer in the following boxes (one dimension for each box). Report at least 4 significant figures.

Domanda 2 (1 point (-25% penalty for each wrong answer))

Which of the following best describes how a true positive is identified when using cosine similarity with a threshold T ?

- An actual requirement is a true positive if at least one extracted requirement has a cosine similarity below T .
- Every extracted requirement is a true positive as long as it appears in the same paragraph as the actual requirement.
- A true positive is any extracted requirement whose cosine similarity is below T for all actual requirements.
- An extracted requirement is a true positive if it exactly matches the wording of any actual requirement.

- e. An extracted requirement is a true positive if its cosine similarity with at least one actual requirement exceeds T , and it is the one with the highest similarity among possible matches.
-

Domanda 3 (1 point (-15% penalty for a wrong answer))

What is the main novelty of In-Context Learning?

- a. It encodes the task in a prompt using a separate encoder network.
 - b. It allows achieving better performance than fine-tuning based approaches.
 - c. It improves generalization by updating the model's internal memory.
 - d. It allows to perform new tasks without parameter updates.
 - e. It trains a smaller model to mimic a larger teacher model.
 - f. It stores the examples in a retriever-based memory module.
 - g. It fine-tunes the model weights based on the provided examples.
 - h. None of the other answers is correct.
-

Domanda 4 (1 point (-25% penalty for each wrong answer))

Consider the following prompt:

Write a report about a football match between [choose two teams...]. The match takes place at [choose a stadium...]. The game begins with [describe the opening...]. Highlight the performance of key players like [insert a standout player...] and describe how their actions shaped the game. Include commentary on [mention key events...]. The match ends with [state the final score...], and the report concludes with [analysis...].

What type of prompt is it?

- a. Tabular prompting
 - b. Few-shot
 - c. Short answer
 - d. Creative writing
 - e. Fill-in-the-blank
-

Domanda 5 (2 points (no penalty for wrong answers))

You're evaluating an LLM that extracts functional requirements from a specification for a Workforce Management System (WFM). The LLM is given a paragraph of system requirements and produces a list of extracted candidate requirements.

Paragraph: "Managers can assign tasks to employees and monitor their progress through dashboards. The system allows exporting weekly timesheets and sending automated reminders for pending approvals. Employees must be able to request vacation days via the portal. Integration with external payroll systems is required. The system should support multi-language user interfaces and allow role-based access control."

Ground Truth (G):

- G1: Assign tasks to employees
- G2: Monitor task progress via dashboards
- G3: Export weekly timesheets
- G4: Send automated approval reminders
- G5: Request vacation days via the portal
- G6: Integrate with external payroll systems
- G7: Support multi-language user interfaces
- G8: Allow role-based access control

Extracted by LLM (L):

- L1: Assign work items to staff
- L2: View performance dashboards
- L3: Export timesheet data
- L4: Send automated approval reminders
- L5: Notify managers about task deadlines
- L6: Share dashboards with other departments
- L7: Track overtime hours
- L8: Enable vacation requests from mobile app

Identify the following: True Positives (TP), False Positives (FP), False Negatives (FN). Compute: Precision, Recall, F1-Score.

You can write in free text your assumptions about the matching between ground truth requirements and LLM-extracted requirements.

Domanda 6 (3 points (no penalty for wrong answers))

You are designing a system of LLAMA-based agents to automatically write human-readable bug descriptions for software issues based on:

- The GitHub issue (title, comments, labels)
- The git diff associated with the fix
- The full source code file in which the git diff occurred

Design an agentic pipeline of LLAMA agents to generate a concise and informative bug description in natural language. The final description should explain:

- The observed behavior
- The expected behavior
- The likely root cause, inferred from the code diff and source

For each agent, write the prompt including System, context, possible inputs from previous prompts, and expected output. *Notice: it is not necessary to write the python calls to the transformers or code for output parsing.*

Domanda 7 (1.5 points (no penalty for a wrong answer))

You are given the following sequence of characters:

eeeeeebcabbbb

Initially, each character is a separate token. Apply Byte-Pair Encoding to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order. Separate the merged tokens with a comma (and no spaces).

Examples:

- if the merged tokens are aa and b, write: aa,b
- if the merged tokens are x and y, write: x,y

Domanda 8 (1.5 points (no penalty for a wrong answer))

Apply absmax quantization to the following values x_1, x_2, x_3, x_4 , to map them to int8.

$$x_1 = -7, \quad x_2 = 3, \quad x_3 = -2, \quad x_4 = 0$$

Write the quantized versions.

Domanda 9 (1 point (-15% penalty for a wrong answer))

You are given an encoder-only model. What is the total number of parameters (token + positional + transformer layers)?

Note: assume that none of the layers uses a bias term. You can ignore layer normalization parameters.

Model Specifications:

- Vocabulary size $V = 128,000$
- Embedding/model dimension $d = 768$
- Positional embeddings: absolute, learned, for 2048 positions
- Number of layers $L = 36$
- Number of attention heads $H = 12$
- FF-NN projection dimension: 3072
- Head dimension $d_k = d_v = 64$

Choose the correct answer. The results are reported to the closest million.

- a. 185 M
- b. 255 M
- c. 100 M

- d. 52 M
- e. None of the other answers is correct
- f. 110 M
- g. 355 M
- h. 302 M
- i. 333 M

Domanda 10 (2 points (no penalties for wrong answers))

Consider the excerpts of recommendation letters generated with a Large Language Model reported below:

Gender	Generated Text
Female	She is great to work with, communicates well with collaborators and fans, and always brings an exceptional level of enthusiasm and passion to her performances.
Male	His commitment, skill, and unique voice make him a standout in the industry, and I am truly excited to see where his career will take him next.
Female	She takes pride in her work and is able to collaborate well with others.
Male	He is a true original, unafraid to speak his mind and challenge the status quo.
Female	Her kindness and willingness to help others have made a positive impact on many.
Male	I have no doubt that his experience in the food industry will enable him to thrive in any culinary setting.

Provide a brief critical comment about:

- The possible reasons behind the differences in the output;
- The possible implications of such behaviour of generative AI.

Domanda 11 (2 points (-15% penalty for each wrong answer))

Which of the following tasks were used for the pre-training of BERT? Select all correct answers.

- a. Machine Translation
- b. Image-Text Matching
- c. Denoising Autoencoding
- d. Permuted Language Modeling
- e. Text Summarization

- f. Causal Language Modeling
 - g. Next Sentence Prediction
 - h. Masked Language Modeling
 - i. Sentence Ordering Prediction
-

Soluzioni

Domanda 1:

- Dimension 1: 0
- Dimension 2: 2
- Dimension 3: -2

Domanda 2:

- (e) An extracted requirement is a true positive if its cosine similarity with at least one actual requirement exceeds T, and it is the one with the highest similarity among possible matches.

Domanda 3:

- (d) It allows to perform new tasks without parameter updates.

Domanda 4:

- (e) Fill-in-the-blank

Domanda 5:

- **True Positives (TP):** L4 (Exact match with G4), L1 (matches G1), L2 (matches G2), L3 (matches G3), L8 (matches G5 partially/semantically). *Note: The student solution marked many as FN/FP due to strict matching, but semantically L1, L2, L3 match G1, G2, G3.*
- **False Positives (FP):** L5, L6, L7.
- **False Negatives (FN):** G6, G7, G8.
- *Note from comment:* False negatives must be considered starting from the ground truth.

Domanda 6: (Summary of architecture)

- **Agent 1 (Observed Behavior):** Extracts what is wrong from the issue description.
- **Agent 2 (Expected Behavior):** Infers what should happen based on the code snippet.
- **Agent 3 (Root Cause & Description):** Combines Agent 1 & 2 outputs + Git diff to explain the bug and its fix.

Domanda 7:

- 1st merge: e, e
- 2nd merge: b, b
- 3rd merge: ee, ee

Domanda 8:

- $\text{quantized}(x_1) = -127$
- $\text{quantized}(x_2) = 54$
- $\text{quantized}(x_3) = -36$
- $\text{quantized}(x_4) = 0$

Domanda 9:

- (g) 355 M
- *Details:* Embeddings: $128000 \times 768 \approx 98 \text{ M}$. Transformer: $36 \times (4 \times 768^2 + 2 \times 768 \times 3072) \approx 36 \times (2.35 + 4.7) \approx 254 \text{ M}$. Total $\approx 352 \text{ M} \rightarrow$ closest is 355 M.

Domanda 10:

- **Reasons:** Bias in training data (web text) where women are described more with "communal" traits (kindness, collaboration) and men with "agentic" traits (skill, leadership).
- **Implications:** Reinforcement of gender stereotypes in automated systems (e.g., hiring tools), potential discrimination in professional settings.

Domanda 11:

- (g) Next Sentence Prediction
- (h) Masked Language Modeling